

Google's Hat Puzzle — Assignment Analysis

ITAI-1370 · History of Artificial Intelligence · Houston Community College
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Task 1 — The Main Puzzle

Puzzle source: puzzles.nigelcoldwell.co.uk/thirtynine.htm

Puzzle Description

One hundred prisoners are each assigned a randomly colored hat — either red or blue. Each prisoner can see all the hats in front of them but cannot see their own. Starting from the back of the line, each prisoner must call out a single color. If they call their own hat color correctly, they survive. The prisoners may agree on a strategy beforehand but may not communicate once the game begins.

The Strategy — Parity Encoding

The optimal strategy relies on a mathematical concept called **parity** — whether a count is odd or even. Before the game, all prisoners agree on one rule:

Person 1 (at the back): Counts all red hats visible in front of them. If the count is odd, they call **red**. If the count is even, they call **blue**. This broadcasts the starting parity to the entire group. Person 1 has a 50% survival chance.

Person 2 onward: Each person tracks how many reds have been called before their turn, counts the reds they can see ahead, and uses the starting parity to deduce their own hat color with 100% accuracy.

The Deduction Formula

Each person from position 2 onward applies this logic:

What I know	How I use it
Starting parity (odd/even) — from Person 1's call	This is my reference point
Reds called before my turn	I tracked every call I heard
Reds I can see ahead of me	I count them directly
Starting parity – reds called – reds ahead = ?	If I need one more red → I'm red. If not → I'm blue.

Result

99 out of 100 prisoners survive — guaranteed. Only Person 1 faces a genuine 50/50 chance. For n prisoners, the survival rate is $(n - \frac{1}{2})/n$, which approaches 100% as the group grows.

10-Person Worked Example

The table below shows the parity deduction for a group of 10 (hats: R R R B R B R R B B):

Person	Odd/Even to start	Reds called before	Reds in front	Says	Survives
1	—	0	5	red	Yes (50/50)
2	odd	1	5	blue	Yes
3	odd	1	4	red	Yes
4	odd	2	3	red	Yes
5	odd	3	2	red	Yes
6	odd	4	2	blue	Yes
7	odd	4	1	red	Yes
8	odd	5	0	red	Yes
9	odd	6	0	blue	Yes
10	odd	6	0	blue	Yes

Task 2 — AI Problem Solving Connections

The Puzzle as an AI Problem

The hat puzzle is a simplified model of the type of problem AI is designed to solve: **making optimal decisions under incomplete information using shared rules and inference**. Each prisoner is essentially an agent that cannot observe its full environment but must still act correctly. This is the foundation of many AI architectures.

Direct Connections to AI Concepts

Parity bits and error correction

Claude Shannon's 1948 information theory paper introduced the concept of encoding one bit of information to detect and correct errors in data transmission. The hat puzzle is a live demonstration of a single parity bit — Person 1's call is that bit, and every subsequent person uses it to error-correct their own guess.

Binary classification

Every person in the puzzle performs a binary classification task: red or blue. AI classifiers (logistic regression, neural networks, SVMs) do exactly this — given observable features, output one of two categories with maximum accuracy.

Bayesian inference

Each person updates their belief about their own hat color as new evidence arrives (each call heard). This is Bayesian reasoning: prior belief + new evidence = updated posterior. AI systems use this in natural language processing, spam filters, and medical diagnosis.

Multi-agent coordination

The prisoners form a cooperative multi-agent system. Modern AI uses multi-agent frameworks (reinforcement learning with shared rewards) to solve similar problems where agents must coordinate using minimal communication.

Spin-off: More colours

The 'More Colours' extension (3+ hat colors) maps directly to multi-class classification in AI. Instead of one parity bit, multiple agents at the back each encode one additional bit — equivalent to using more output nodes in a neural network classifier.

Task 3 — Reflection Talking Points

Use these bullet points to write your own reflective narrative in your own words:

- My first instinct was that the puzzle was unsolvable — you can't know what you can't see. That assumption turned out to be the obstacle.
- The breakthrough was realizing one person doesn't need to save themselves — they need to give everyone else what they need to save themselves.
- Parity felt abstract at first. Once I saw it as 'odd/even is the only thing that needs to be communicated,' it clicked — one bit of shared information was enough for 99% survival.
- Tracking the running count of reds called felt like keeping score in a game — each call updated my mental model of what the group looked like.
- The connection to AI surprised me: this puzzle is a physical demonstration of error-correcting codes, which are in every device I use daily.
- The 'More Colours' extension showed me how you scale the strategy — more bits needed means more people at the back sacrifice their certainty to give the group more information.
- The biggest lesson: in AI and in life, optimal decisions don't require complete information — they require the right shared framework and the discipline to apply it consistently.